

Domain Testing on Amazon website

Introduction

- This presentation covers a software testing technique called DOMAIN TESTING on the popular e-commerce platform called Amazon.com to assess its reliability.
- We'll explore 6 test cases, provide their descriptions, define a set of inputs, the expected outputs, and then analyse the actual outputs given by the website.
- This will allow us to understand the importance of domain testing to ensure the reliability of software systems

What is Domain Testing?

Domain Testing is a black-box testing technique that focuses on testing the boundaries and partitions of input domains. It aims to:

- Identify errors in input processing
- Ensure the system handles boundary values correctly
- Verify behavior with valid and invalid inputs
- Test extreme cases and edge conditions

Domain Testing is particularly useful for systems with complex input requirements, like search functions or form submissions.

Domain Testing: Test Case 1 - Search Function

Feature: Amazon's product search function

Test Description: Test various input types in the search bar

Test Cases:

1. Valid input: "wireless headphones"
2. Boundary case: 256-character search string (maximum allowed)
3. Invalid input: Empty search string
4. Edge case: Search with special characters "headphones@#\$\$%"

Expected Results:

- | | |
|------------------------------------|---|
| 1. Display relevant search results | 3. Display an error message |
| 2. Accept input and perform search | 4. Handle or remove special characters and perform search |

Actual Results:

1. Showed results for wireless headphones
2. Shows result relevant to keywords (string used = 256 characters)
3. Goes to Home Page
4. Ignores the special characters and shows relevant results

Domain Testing: Test Case 2 - Price Filter

Feature: Price range filter in search results

Test Description: Test various price range inputs

Test Cases:

1. Valid input: \$10 - \$100
2. Boundary case: \$0 - \$10000 (assuming this is the max range)
3. Invalid input: \$100 - \$10 (min > max)
4. Edge case: \$0.01 - \$0.02 (minimum possible range)

Expected Results:

- | | |
|--|---|
| 1. Display products within the specified range | 3. Display an error message or swap the values |
| 2. Display all products up to \$10000 | 4. Display products (if any) within this tiny range |

Actual Results:

1. Filter Successful, shows expected results
2. Filter Successful, shows expected results
3. Unable to provide such an input
4. Unable to provide input. Minimum value to be given on filter varies with different products.

Domain Testing: Test Case 3 - User Registration

Feature: User registration form

Test Description: Test various inputs for the email field

Test Cases:

1. Valid input: "user@example.com"
2. Invalid input: "notanemail"
3. Edge case: "user+test@example.co.uk" (valid but complex)

Expected Results:

1. Accept the email and proceed
2. Display an error message about invalid email format
3. Accept the email (it's a valid format)

Actual Results:

1. Test Successful, shows expected results
2. Test Successful, shows error message for invalid email format
3. Test Successful, accepts the email as valid format

Domain Testing: Test Case 4 - Product Quantity Selection

Feature: Product quantity selector on product pages

Test Description: Test various quantity inputs

Test Cases:

1. Valid input: 5 (assuming product is in stock)
2. Invalid input: 0

Expected Results:

1. Update quantity to 5
2. Display error or default to minimum quantity (1)

Actual Results:

1. Test Successful, Stock updated successfully
2. Test Failed, cannot input value.

Domain Testing: Test Case 5 - Product Reviews

Feature: Amazon's product review system

Test Description: Test various scenarios for submitting and displaying product reviews

Test Cases:

1. Valid input: Submit a 5-star review with 500-character text
2. Boundary case: Submit a review with the maximum allowed character count
3. Invalid input: Submit a review without selecting a star rating
4. Edge case: Submit a review with only spaces in the text field

Expected Results:

- | | |
|---|---|
| 1. Review successful and displayed on product page | 3. System displays error and prevents submission |
| 2. Review is accepted and posted without truncation | 4. System trims the spaces or prompts for valid input |

Actual Results:

1. Test Successful, Review accepted
2. Test Successful, Review accepted (5000 characters)
3. Test Failed, review submission not successful, system does not allow to click on submit button
4. Test Successful, system accepts the blank review

Domain Testing: Test Case 6 - Shopping Cart

Feature: Amazon's shopping cart functionality

Test Description: Test various operations on the shopping cart

Test Cases:

1. Valid input: Add 10 different items to the cart
2. Boundary case: Add more than maximum allowed number of items to the cart (50 unique items)
3. Invalid input: Attempt to add a negative quantity of an item
4. Edge case: Add an item to the cart, then delete the item from Amazon's catalog

Expected Results:

1. All 10 items are added and displayed
2. System accepts up to the maximum items, then prevents further additions
3. System displays an error message
4. Item is removed from the cart

Actual Results:

1. Test Successful, Products added successfully
2. Test Failed, unable to add items
3. Test Failed, unable to input a negative value
4. Test Successful, item deleted successfully

Domain Testing: Test Case 7 - Address Input

Feature: Amazon's address input for shipping

Test Description: Test various inputs for the shipping address form

Test Cases:

1. Valid input: Enter a complete US address with all fields filled
2. Boundary case: Enter the maximum allowed character count for each field
3. Edge case: Enter an address with non-ASCII characters (e.g., ñ, é, ß)

Expected Results:

1. Address is accepted and saved successfully
2. System accepts input up to maximum length for each field
3. System either accepts and properly stores the characters, or provides clear guidance on allowed characters

Actual Results:

1. Test Successfully executed
2. Test Successfully executed
3. Test Failed, error in address field

Conclusion

In this presentation, we explored how domain testing can be applied to thoroughly test and verify the working of a software, in our case Amazon website).

We were able to create targeted test cases for different functionalities of the website. This approach ensures that each feature is rigorously examined for potential issues.

Ultimately, domain testing helps in delivering a more reliable and user-friendly software product.